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CONTRIBUTION TO ACCURATE MEASUREMENT OF AERODYNAMIC DRAG ON A MOVING VEHICLE FROM COAST-DOWN TESTS AND DETERMINATION OF ACTUAL ROLLING RESISTANCE*

G. ROUSSILLON

Société des Automobiles Peugeot, Centre d'Etudes de Paris, Secteur Etudes Aérodynamiques, 18, rue des Fauvelles, 92250 La Garenne Colombes (France)

Summary

A realistic measure of C_D is obtained from deceleration tests in light cross-wind on a track having slight gradient by considering velocities relative to the track and to the wind. Combination of this measure with C_D obtained in wind-tunnel tests and with a value for the base rolling resistance determined at low speed on a gentle incline permits determination of a realistic value for the rolling resistance for different types of tyre and road surface. It is possible to establish energy balances and correlations between wind-tunnel, track and roller bench measurements.

Notation

M	vehicle mass
I	inertial moment of rotating parts
ΔM_i	correction to mass of vehicle to account for addition of rotational to translational kinetic energy
R	rolling radius
V_∞	freestream velocity
V_0	stream velocity in front of vehicle
V_r	track velocity
V_j	wind velocity
ρ	mass density of air
q_∞	freestream dynamic pressure
q_0	measured dynamic pressure in front of vehicle (Prandtl antenna)
K_1	blocking coefficient
K_2	speed coefficient relative to wind
K_r	rolling-resistance coefficient
F_g	global translation inertia
R_ϵ	basic rolling resistance

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$R_{(r+t)}$ rolling resistance and transmission resistance
D aerodynamic drag
Y aerodynamic side-force
L aerodynamic lift
C_D drag coefficient
C_Y side-force coefficient
C_L lift coefficient
A frontal area of vehicle
β yaw angle
α gradient of track
g gravitational acceleration

Subscripts
∞ freestream measurements
0 measurement at Prandtl antenna
r track
F front
R rear
t transmission

I. Coast-down tests

I.1. Introduction

The determination of vehicle drag using a deceleration method was first described by Hoerner [1] in 1935. This pioneer pointed out that the procedure can be performed only in the absence of any wind and on a perfectly horizontal track.

The originality of the procedure here consists in the ability to exploit measurements obtained in the presence of a slight wind making a small angle with the track. The latter may in addition have a slight gradient. Use of the method is then possible only by recording simultaneously both the speed relative to the track and the speed relative to the wind.

The method can be improved by digitizing the analog parameters. In the present experiments, a rate of 250 measurements per second was achieved (see Fig. 1).

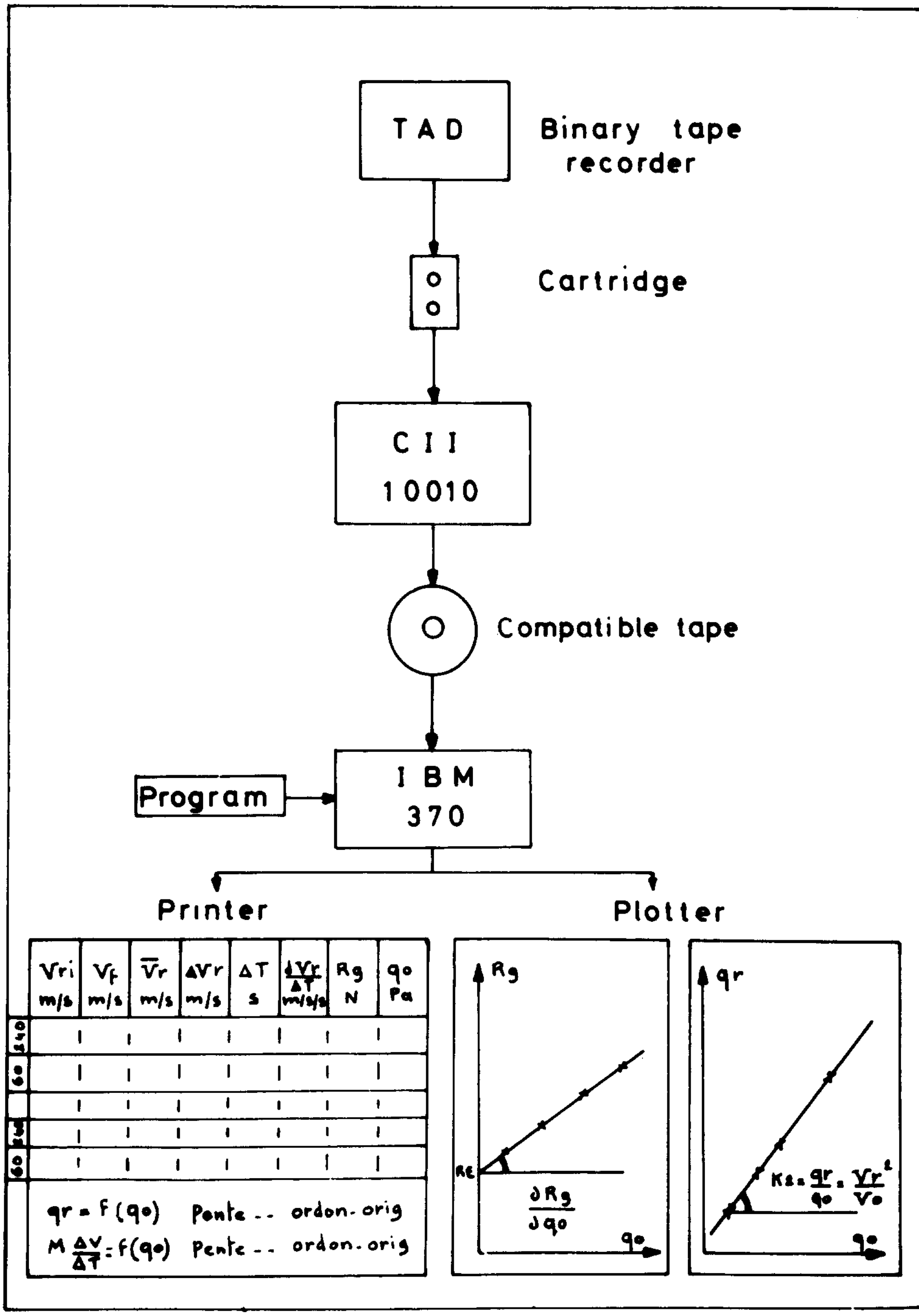

Fig. 1 (a). Schematic diagram of signal treatment.

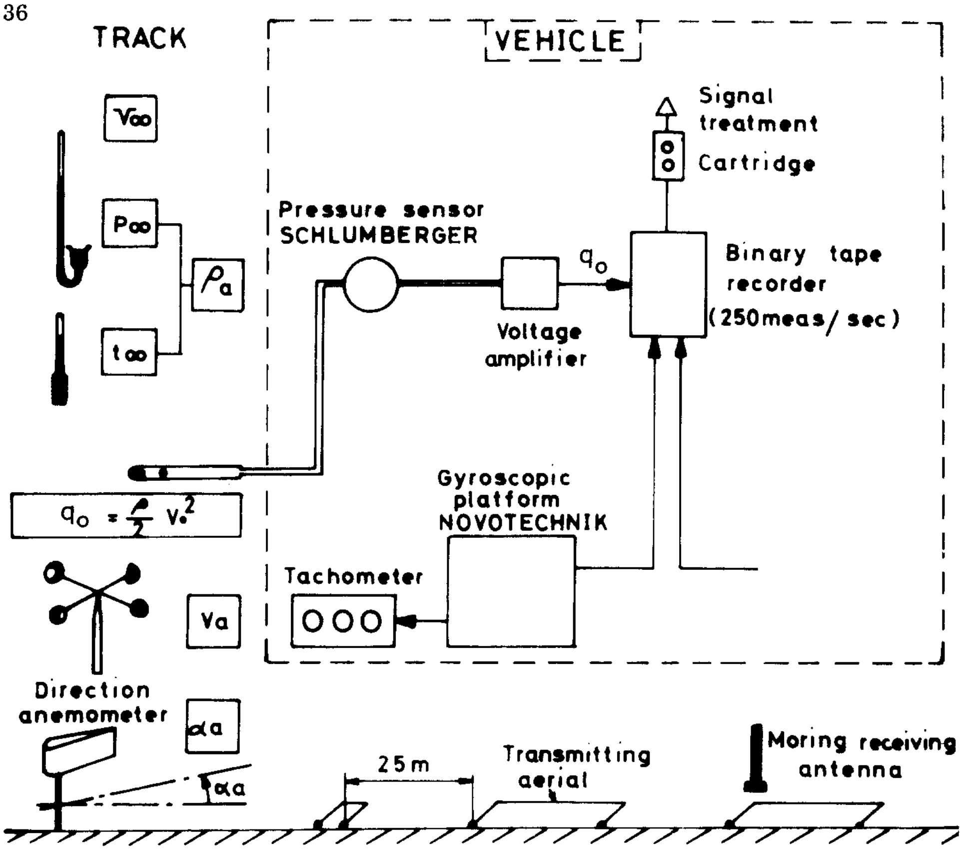

Fig. 1 (b). Schematic diagram of measuring apparatus.

I.2. Measurements

The vehicle (a Peugeot 305 sedan) is launched on a flat track (see Fig. 2) at increasing speeds ranging from 60 to 120 km h^{-1}. The driver then sets the gearbox in neutral. From this time, the vehicle undergoes deceleration. The vehicle's forward momentum comprises two terms:

(1) translational momentum, $M(\mathrm{d}V_r/\mathrm{d}t)$; and

(2) angular momentum from the rotating wheels, $(I/R^2)\cdot(\mathrm{d}V_r/\mathrm{d}t) = \Delta M_i(\mathrm{d}V_r/\mathrm{d}t)$;

to which the following resistances are opposed:

(1) aerodynamic drag, $D = (1/2)\rho_\infty V_\infty^2 C_D A$;

(2) rolling resistance of the tyres and transmission resistance, $R_{(r+t)}$: and

(3) for a sloping track, the gravity component, $\pm Mg \sin\alpha$.

The equation of motion is then expressed according to Newton's second law:

$$(M + \Delta M_i)(\mathrm{d}V_r/\mathrm{d}t) = \underbrace{(1/2)\rho_\infty V_\infty^2 A C_D}_{\text{aerodynamic drag}} + \underbrace{R_{(r+t)}}_{\text{rolling resistance}} \pm \underbrace{Mg\sin\alpha}_{\text{gravity component}} \qquad (1)$$

Fig. 2. *Top*, track at Senonches; *centre*, roller bench; *bottom*, wind tunnel (St. Cyr Institut Aérotechnique).

Three different velocities were considered in the measurements:
(1) V_r, the track velocity;
(2) V_0, the stream velocity in front of the vehicle measured in terms of dynamic pressure using a Prandtl antenna, according to $q_0 = \rho_0(V_0^2/2)$; and
(3) V_∞, the freestream velocity, where $q_\infty = \rho_\infty(V_\infty^2/2)$; and the following basic experimental coefficients were derived:
$K_1 = V_0^2/V_\infty^2 = q_0/q_\infty$, measured in the wind tunnel; and
$K_2 = V_r^2/V_0^2 = q_r/q_0$, measured on the track.

In order to use these measurements to derive aerodynamic drag coefficients, it is necessary to make use of the rolling resistance law, using results obtained on a roller bench (see Figs. 3 and 4):

$$R_{(r+t)} = R_\epsilon + K_r V_r^2 \tag{2}$$

for $10 < V_r < 120$ km h^{-1}.

This may be introduced into the equation of motion (1) to give

$$(M + \Delta M_i)\mathrm{d}V_r/\mathrm{d}t = A C_D \rho_\infty(V^2/2) + (R_\epsilon + K_r V_r^2) \pm Mg \sin\alpha \tag{3}$$

If we write

$$(M + \Delta M_i)\, \mathrm{d}V_r/\mathrm{d}t = F_g \tag{4}$$

and take as single variable the square of the velocity V_0 recorded at the Prandtl antenna in front of the vehicle:

$$q_0 = \rho_0 V_0^2/2 \quad \rightarrow \quad V_0^2 \text{ (with } \rho_0 \simeq \rho_\infty) \tag{5}$$

then by using the velocity equations, we have

$$F_g = A C_D(\rho_0/2)(V_0^2/K_1) + (R_\epsilon + K K_2 V_0^2) \pm Mg \sin\alpha$$

By taking the derivative of this equation with respect to V_0^2, constants such

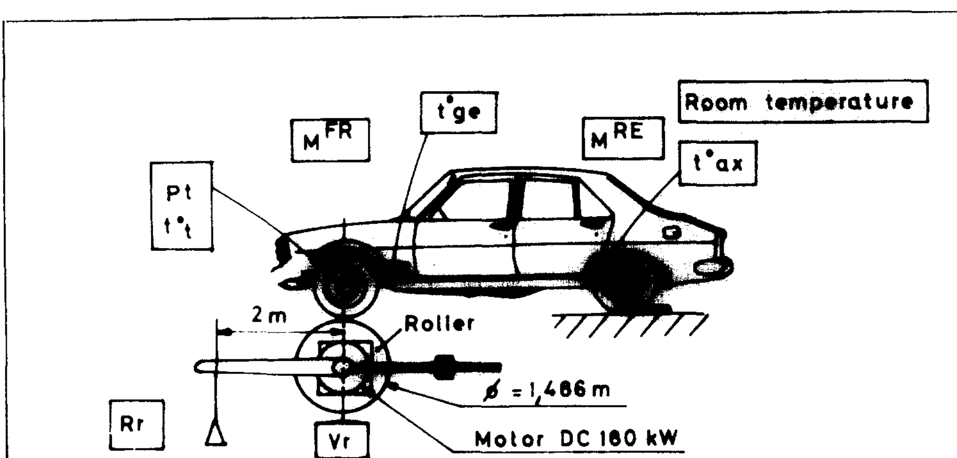

Fig. 3. Schematic diagram of roller bench (Centre d'Etudes de Paris, Société des Automobiles Peugeot).

as the basic rolling resistance and the gravity component may be eliminated, giving

$$\partial F_g/A(\rho_0/K_1)(\partial V_0^2/2) = C_D + [K_1K_2/(\rho_0/2)A]K_r = C_{D(b)} \tag{6}$$

The global resistance expressed in terms of the dimensionless drag coefficient $C_{D(b)}$ is equivalent to the sum of the aerodynamic drag coefficient C_D and a rolling drag coefficient: hence

$$C_D = (K_1/S)(\partial F_g/\partial q_0) - (2K_1K_2/\rho_0 A)K_r \tag{7}$$

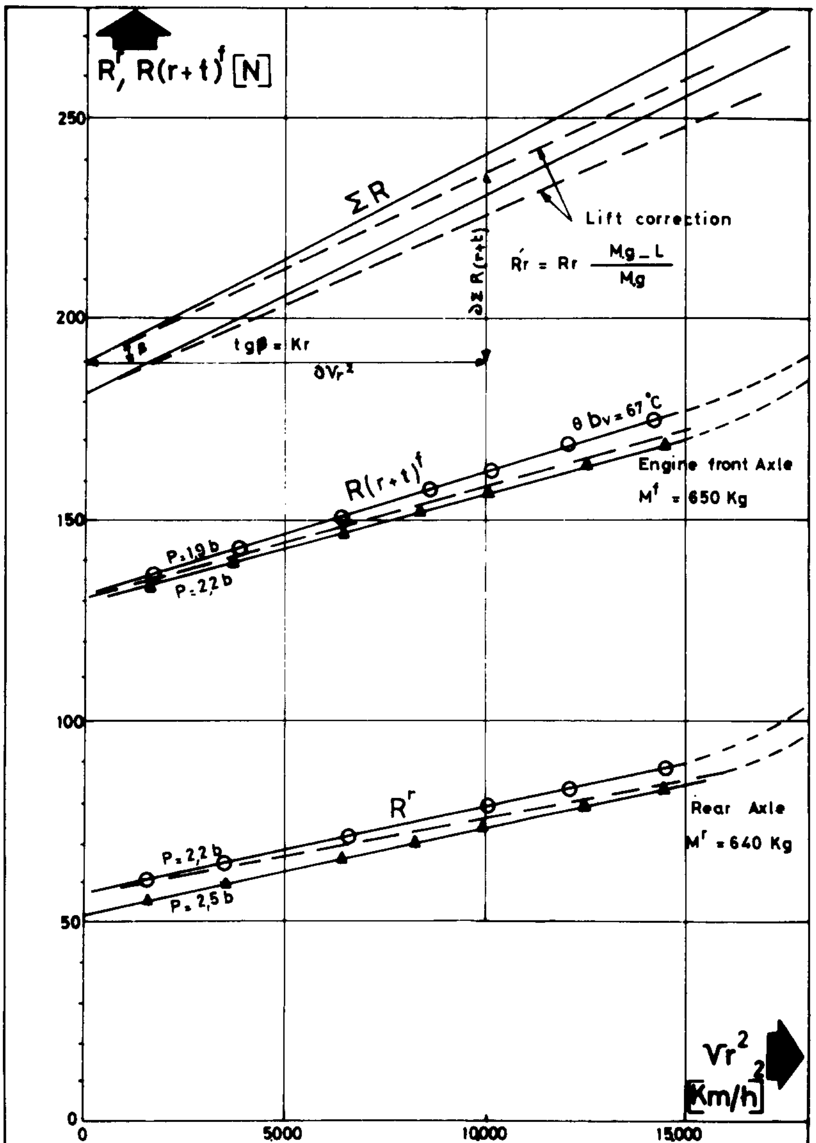

Fig. 4. Determination of K_r using data obtained on the roller bench (see Fig. 3).

TABLE 1

Measurement of C_D by deceleration method for Peugeot 305: experimental data

Run	Direction [a]	V_{init} (m s^{-1})	V_{fin} (m s^{-1})	V_{av} (m s^{-1})	ΔV (m s^{-1})	Duration (s)	dV/dt (m s^{-2})	$F_g(N) = R_g$	PDO(PA) = q_0
3 1	240	16.175	14.369	15.272	1.806	6.410	0.282	376.220	125.614
3 2	60	16.464	15.091	15.778	1.374	6.150	0.223	298.166	153.111
3 3	240	19.081	17.237	18.159	1.844	5.380	0.343	457.567	204.394
3 4	60	19.127	17.698	18.413	1.429	5.230	0.273	364.659	239.831
3 5	240	21.783	20.059	20.921	1.724	4.670	0.369	492.739	249.665
3 6	60	21.403	19.970	20.686	1.432	4.680	0.306	408.628	288.825
3 7	240	23.577	21.869	22.723	1.708	4.320	0.395	527.685	297.541
3 8	60	22.987	21.536	22.262	1.451	4.340	0.334	446.250	349.744
3 9	240	26.050	24.345	25.198	1.705	3.880	0.439	586.596	360.104
310	60	25.814	24.313	25.063	1.501	3.860	0.389	519.263	427.366
311	240	27.111	25.304	26.207	1.807	3.440	0.525	701.248	469.358
312	60	28.667	27.106	27.886	1.560	3.470	0.450	600.331	504.810
313	60	30.973	29.353	30.163	1.619	3.200	0.506	675.625	596.410
314	240	34.539	32.650	33.595	1.889	2.890	0.654	872.618	643.007
315	60	33.127	31.398	32.263	1.729	2.980	0.580	774.414	694.812

Direction of run [a]		*Slope*	*Intersection with ordinate*
240	PDR(PDO) [b]	1.059	0.0 (PA)
240	FMG(PDO)	0.952 (N/PA)	280.8 (N)
240	FVT(PDO)	0.955 (N/PA)	253.1 (N)
60	PDR(PDO)	0.952	−5.9 (PA)
60	FMG(PDO)	0.880 (N/PA)	183.6 (N)
60	FVT(PDO)	0.881 (N/PA)	152.5 (N)

[a] See diagram bottom right.

[b] PDR = $q_r = \rho V_r^2/2$; FMG = $F_g = (M + \Delta M_i)\gamma$ (γ measured); FVT = $F_g = (M + \Delta M_i) dV/dt$.

I.3. Results

The results of the tests on a Peugeot 305 sedan car are presented in Fig. 4 and Tables 1 and 2.

The agreement between the value C_D = 0.425 measured on the track with that obtained in the wind tunnel (C_D = 0.435) is satisfactory (see below and Fig. 5). The main interest of the track deceleration method is that

TABLE 2

Measurement of C_D by deceleration method for Peugeot 305: results

Gross C_D	*Definition*	*Track to 240*[a]		*Track to 60*
	$C_D(b) = (\partial D/\rho q)(K/A)$	$\frac{0.995 \times 0.93}{1.75}$ = 0.507		$\frac{0.881 \times 0.93}{1.75}$ = 0.468
Wind correction	$K_1 = q_r/q_0 = V_r^2/V_0^2$	1.059		0.952
V_r^2 term rolling resistance (roller bench)	$K_r = \partial R_r/\partial V_r^2$ $\sum K_r$	front axle	5.6×10^{-3}	rear axle
Mean deviation	$\bar{\beta} = \arctan g(V_j/V_\infty)$	0		0
Lift at V_{max}	$L^{F \cdot R} = A\, C_L^{F \cdot R} q_0/K$ $(L^F + L^R)$	front axle 236	411	rear axle 175
Lift correction	$(Mg - L)/Mg = \rho_a$	front axle 0.965		rear axle 0.973
V_r^2 term corrected	$K_r^1 = \rho_a K_r$	front axle	5.45×10^{-3}	rear axle
Gross basic rolling resistance	$R_{\epsilon(b)} = F_G(q_0 = 0)$	253	1	152
Gravity component	$Mg \sin \alpha$	−47		+47
Mean basic rolling resistance	$R_\epsilon = R_{\epsilon(b)} \pm Mg \sin \alpha$	*206*		*199*
Pseudo-C_D of rolling	$C_D = (K_1/A)[(2/\rho)K_r K_2]$	0.065		0.058
Skid correction	$\partial C_D = \beta(\partial C_D/\partial \beta)$	0		0
C_D ($\beta = 0°$)	$C_{D(b)} - C_{x(r)} + C_{D(b)}$	0.442		0.41
C_D (mean)	$\frac{1}{2}[C_D(240) + C_D(60)]$		0.425	

[a]Refer to diagram in Table 1.

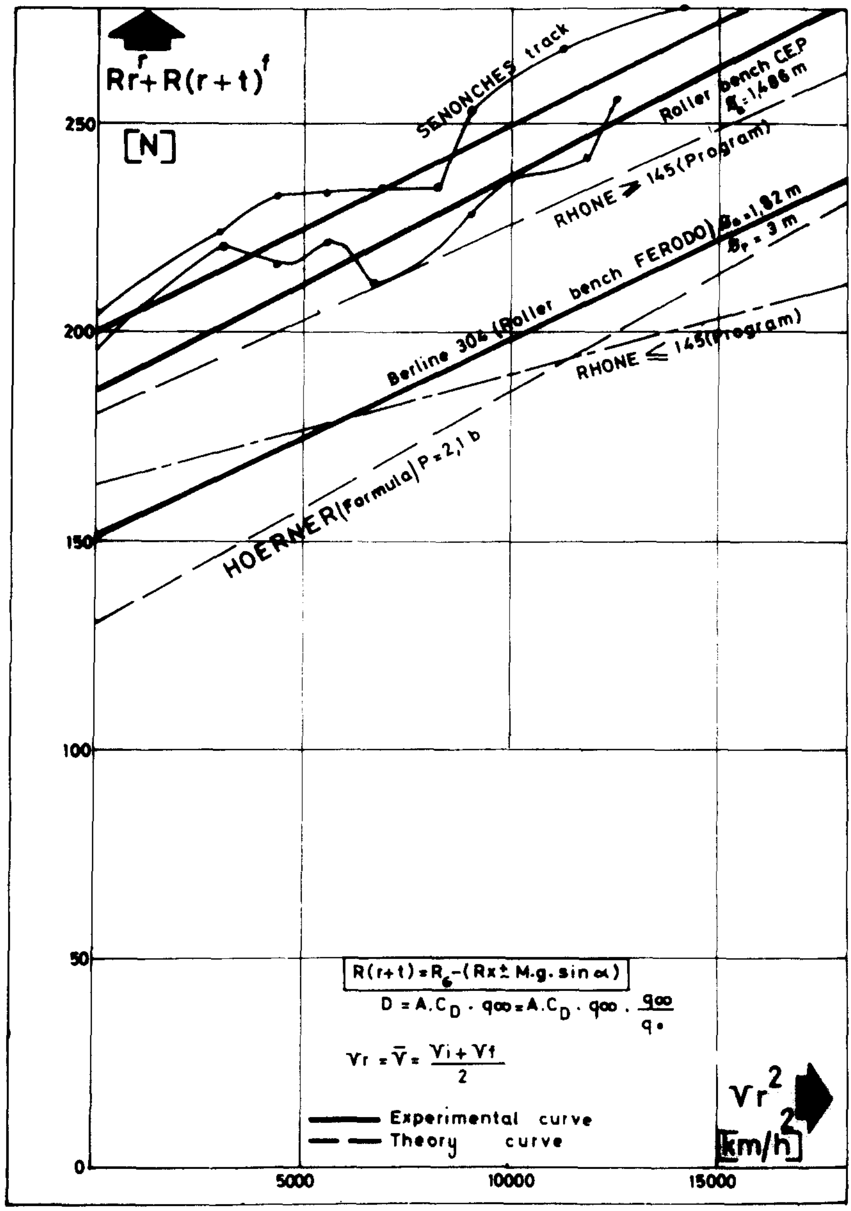

Fig. 5. Summary of measurements of rolling resistance.

it supplies a C_D value appropriate to actual conditions of use; the wheels are rotating, the relative motions of the road, vehicle and air stream (null-limit layer) are preserved, and there are no blocking or wall effects since operation is effectively in infinite space.

Turbulence and the unstationary nature of natural wind are taken into ac count, while in the wind tunnel the turbulence is artificially regulated and the flow is stationary. On the other hand, in track tests it is necessary to proceed with a slight wind ($V < 4$ m s^{-1}) and a low angle with the vehicle axle ($< 7°$).

An interesting development of the deceleration method would consist in the simultaneous measurement of the lift variation on the front and rear axles in order to make comparisons with the results obtained in the wind tunnel, where because the limit conditions with respect to rotating wheels and relative vehicle, airstream and road velocities are not fulfilled, there remains doubt concerning the absolute lift values.

II. Determination of actual rolling resistance

II.1. Introduction

Following the development of the deceleration method, we have carried out wind-tunnel tests on vehicles at the St. Cyr Institut Aérotechnique, with calibration obtained by correlation between road tests and maximum-speed tests.

The aerodynamic measurements performed in the wind tunnel, allowing calculation of the coefficients C_D (drag), C_L for the lift on the front and rear axles, and C_Y for the lateral forces due to cross-wind for $\beta \leqslant 10°$, indicated a lack of conformity in the two test methods concerning the basic resistance at zero speed in the track and roller-bench tests. This did not prevent determination of C_D but led us to introduce an intermediate test, as follows. This test consists in using a track with a known gradient, over which the vehicle accelerates but reaches a sufficiently low speed that aerodynamic resistance may be disregarded. The basic tyre resistance is also determined more precisely and the rolling resistance may be determined from the difference between the global resistance (deceleration on track) and the drag (wind tunnel).

The combination of these three test methods allows measurement of the rolling resistance under real conditions of use and subsequent study of the influence of such parameters as tyre inflation pressure, adjustment angles, nature of the road surface, and of different types of tyres.

II.2. Measurements

II.2.1 Determination of global resistance

The global resistance R_g is determined by measurements of deceleration on a track as described in Section I above.

II.2.2 Determination of basic rolling resistance

The vehicle is launched with zero initial velocity on an inclined plane whose gradient is known with precision (see Fig. 6). It is subject to a driv-

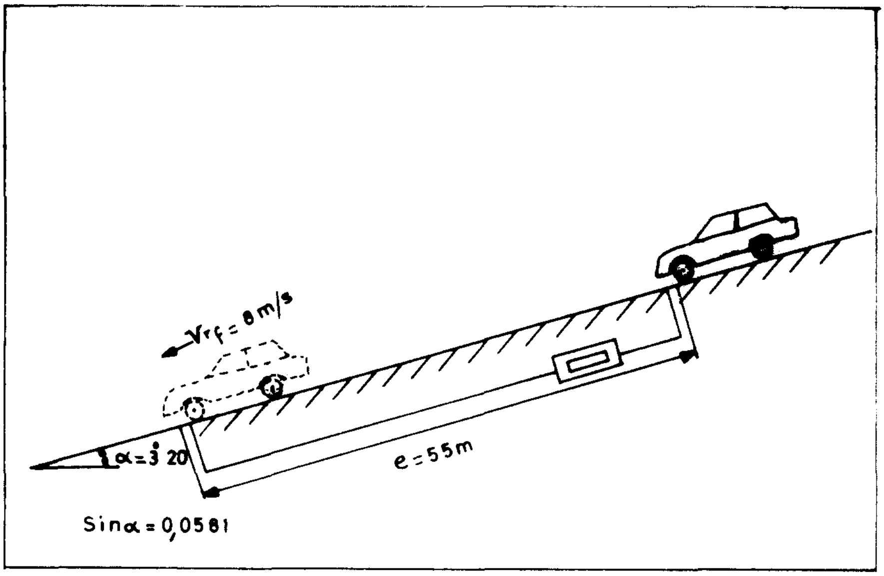

Fig. 6. Schematic illustration of slope tests.

ing force due to the gravity component, and to the resistance forces:

(1) rolling resistance, $R_r = R_\epsilon + k_r V_r^2$;

(2) aerodynamic resistance, $D = A C_D \rho_\infty (V_\infty^2/2)$; and

(3) wheel rotation inertia, $R_i = I\omega^2 = (I/R^2)(\mathrm{d}V/\mathrm{d}t) = \Delta M_i(\mathrm{d}V/\mathrm{d}t)$.

The speed reached being low (< 9 m s^{-1}), the quadratic terms (for aerodynamics and rolling) can be neglected. The simplified equation of motion is thus

$$Mg \sin\alpha - R_\epsilon = (M + \Delta M_i)(\mathrm{d}V_r/\mathrm{d}t) \tag{8}$$

Measurement of the time t necessary for the vehicle to cover the distance e ($\sim$16 s) allows calculation of the mean acceleration according to

$$\mathrm{d}V_r/\mathrm{d}t = 2e/t^2 \tag{9}$$

and of the basic rolling resistance:

$$R_\epsilon = Mg \sin\alpha - (M + \Delta M_i)(2e/t^2) \tag{10}$$

II.2.3 Determination of aerodynamic characteristics

The vehicle to be tested was placed in the 15 m^2 working section (having slotted walls) of the St. Cyr Institut Aérotechnique wind tunnel. The use of slotted walls gives results equivalent to those for a conventional 45 m^2 working section, thus reducing the blockage correction to 2%.

A six-component balance (Fig. 7) allows the three forces (drag D, lift L, side force Y) and the three moments (rolling, pitching, yawing) to be measured, and thus the forces acting on each axle to be plotted.

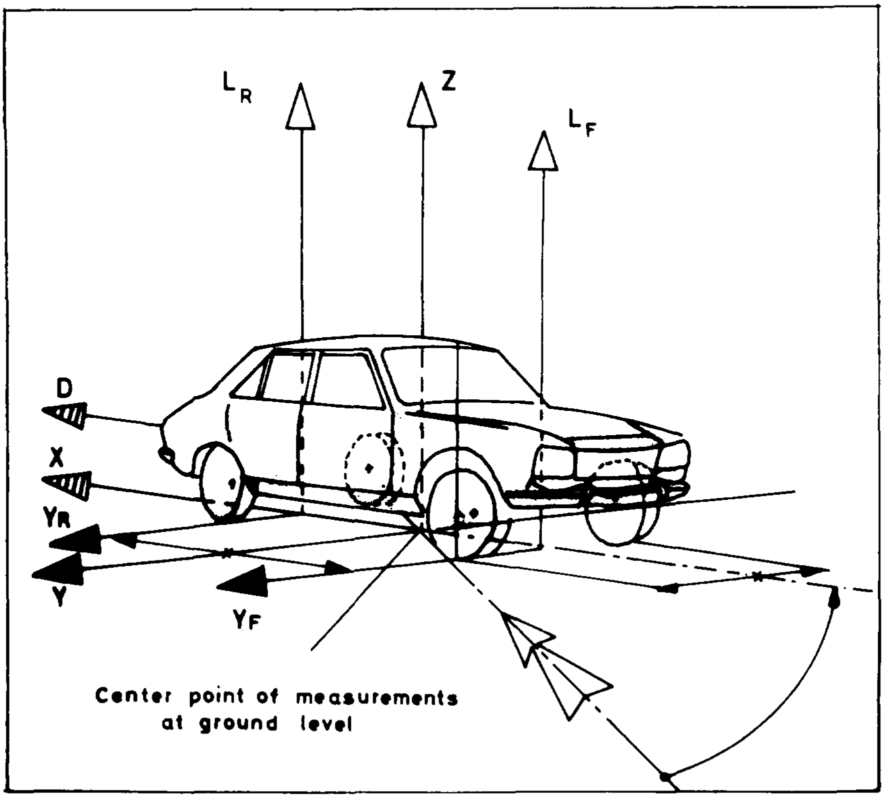

Fig. 7. Wind-tunnel force balance.

In making use of these measurements to obtain aerodynamic coefficients, again the same difficulty as in the track test method is encountered, namely, that the resistance forces of aerodynamic origin are dependent on the vehicle speed relative to the wind (V_0, V_∞), whereas the rolling resistance forces depend mainly on the speed relative to the road, V_r, and only partly on the speed relative to the wind (ventilation in wheel housings, thermal exchanges, lightening due to lift, yaw angle).

For each test it is possible to calculate

$$R_{(r+t)} = R_g - D \pm Mg \sin \alpha$$

with $R_g = f(V_r^2)$ and $D = \varphi(q_\infty) = AC_D q_0/K_1$, from which $R_{(r+t)} = f(V_r^2)$ may be expressed.

If it is required to eliminate the resistance of the transmission it is necessary to uncouple the half transmission-shafts and launch the vehicle with the help of a "pusher" that is released before the vehicle enters the measurement area, in order not to introduce aerodynamic interaction.

II.3. Results

II.3.1 Rolling resistance deduced from deceleration tests

The basic rolling resistance referred to the zero ordinate (null speed) of the recording representation was obtained (Table 2) as $(240)R_\epsilon$ = 206 N and $(60)R_\epsilon$ = 199 N, yielding a mean value $\overline{R}_\epsilon$ = 202 N (term coefficient V_r^2).

Linear regression of points obtained by differences of global resistance, drag and gravity components gives a straight line in V_r whose directing coefficient is very close to that determined on the roller bench (see Fig. 4):

on track $K_r = 5.5 \times 10^{-3}$ N $(\text{km h}^{-1})^{-2}$

on bench $K_r = 5.6 \times 10^{-3}$ N $(\text{km h}^{-1})^{-2}$

II.3.2 Basic resistance measured on inclined plane

The essential parameters M = 1300 kg, ΔM_i = 40 kg, e = 55 m, and $\sin \alpha$ = 0.0581, combined with the average t-value from five measurements, $\bar{t}$ = 16.1 s, yield $\overline{R}_\epsilon$ = 170 N.

There is here a divergence of 15% relative to the result obtained on the Senonches track. Part of this divergence is due to the fact that the inclined plane has low roughness while the deceleration track is a concrete surface with higher roughness.

II.3.3 Basic resistance deduced from roller-bench tests

The extrapolated zero ordinate of the curve (see Fig. 4)

$$R_i^r + R_{(r+t)}^F = f(V_r^2)$$

gives the value R_ϵ = 185 N. This value, which is between the two preceding ones, can be explained in terms of the intermediate roughness of the roller surfaces with respect to the inclined plane and the deceleration track, since the former were covered with emery paper.

III. Conclusions

Free-wheel deceleration tests may serve to determine the rolling resistance, or more exactly drag resistance, of tyres under actual conditions of use, i.e. such that the geometric adjustment of the wheel-trains, losses due to ventilation in the wheel housings, thermal exchanges with the moving air as well as with the track, variable apparent weight with aerodynamic lift, and track granulometry may be correctly taken into account.

Acceleration tests using an inclined plane are easy to perform and give a basic resistance value with good definition, since a 0.3 N m^{-2} variation of inflation pressure results in a 0.3 s time variation and a 10 N resistance variation, that is, 5%.

Since the basic resistance value recorded on two different road surfaces in the present study shows significant variation, it would be interesting to

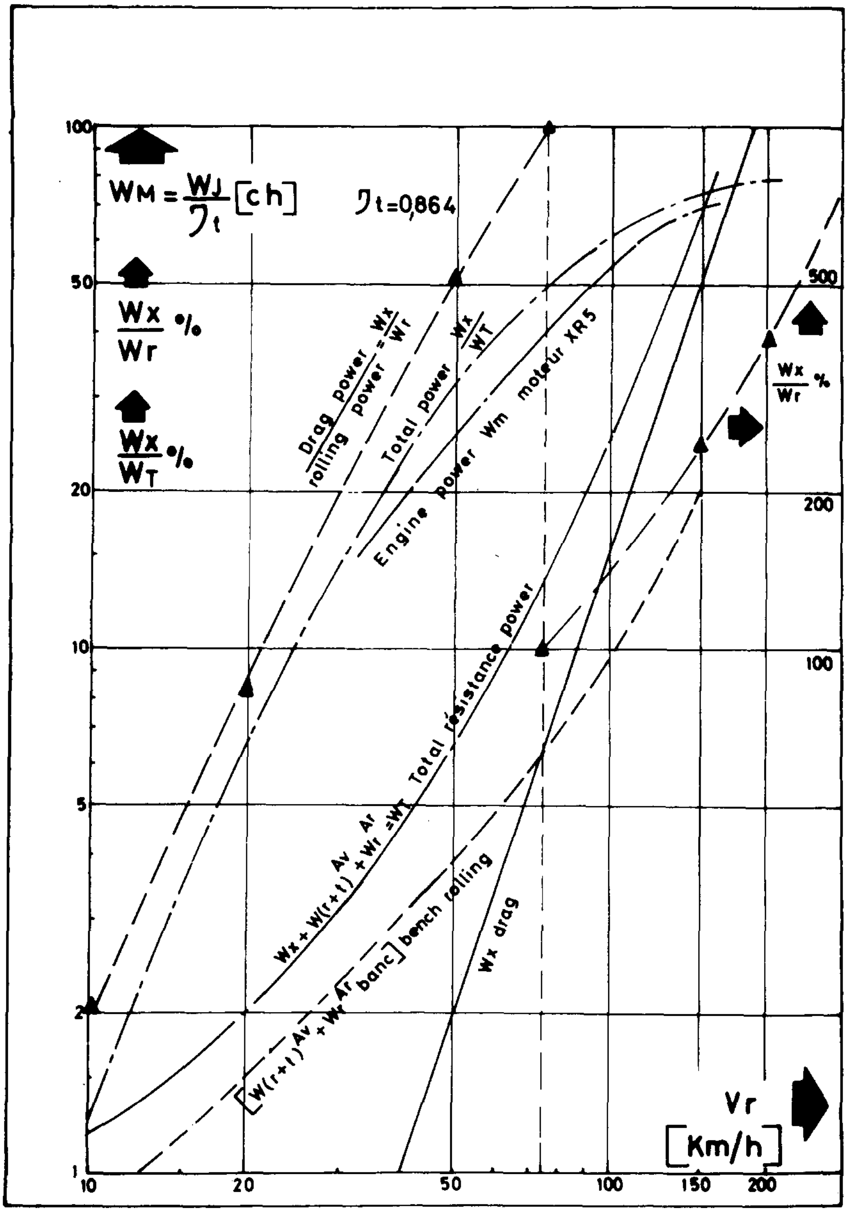

Fig. 8. Variation of critical parameters with road velocity.

continue these tests by performing systematic measurements in order to quantify the effects of different types of tyres or road surfaces.

The combination of track deceleration, inclined-plane tests and measurements carried out in an artificial environment (wind-tunnel and roller-bench tests) allows:

(1) thorough examination of the problems relating to the analysis of energies absorbed in aerodynamic and rolling resistance under actual condi tions of use (Fig. 8);

(2) correlations to be obtained between the track and roller-bench or wind-tunnel results, allowing adjustment of the wind tunnel (correction of blockage effects) and regulation of the energy absorbed using a roller bench, in order to carry out consumption tests, maximum-speed tests, and engine-cooling tests in a satisfactory manner (see Fig. 8).

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